# The Virasoro spectral $R$-matrix: closed form and non-formality

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**Abstract**

The spectral $R$-matrix of the Virasoro algebra on a primary state of conformal weight $h$ admits the closed form $R(z) = z^{2h}\exp(-c/(4z^2))$. Only even powers of $1/z$ appear, reflecting bosonic parity. The non-terminating character of the exponential series is the genus-0 signature of shadow class $M$: unlike Heisenberg ($R(z) = z^k$, polynomial) or affine Kac–Moody ($R(z) = z^{\Omega/(k+h^\vee)}$, terminating), the Virasoro $R$-matrix has infinitely many corrections in $1/z^2$. The cubic shadow coefficient $S_3(\mathrm{Vir}_c) = 2$ is independent of the central charge $c$, providing a finite algebraic proof of class $M$ membership. The first correction $R_2 = -c/4$ is the genus-0 non-formality witness from which the infinite primitive-cumulant series descends.

## 1 Introduction

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve $X$. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra. The binary genus-0 collision residue

$$r(z) := \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \tag{1.1}$$

is the spectral $r$-matrix of $\mathcal{A}$. The collision residue lives on the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the $E_1$ convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$ that retains the full spectral-parameter dependence. The modular characteristic $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ is its $\Sigma_2$-coinvariant.

The $r$-matrix encodes the leading singularity of the bar differential and controls the commuting Hamiltonians of the shadow connection at genus 0. On a rank-1 primary line, the path-ordered exponential of $r(z)$ produces the spectral $R$-matrix, which is a scalar function of the spectral parameter $z$.

For the Virasoro algebra $\mathrm{Vir}_c$, the $R$-matrix on a primary state of weight $h$ closes into an elementary function. This paper derives the closed form, expands it in series, extracts the shadow coefficients, and uses the result to distinguish the Virasoro from simpler families at the genus-0 level.

The passage from OPE to $r$-matrix uses the $d\log$ absorption principle: the bar propagator $d\log(z-w) = dz/(z-w)$ absorbs one power of the OPE singularity, so the collision residue has pole orders one less than the OPE. The Virasoro OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \tag{1.2}$$

has poles at orders $4, 2, 1$. After $d\log$ absorption, the collision residue carries poles at orders $3, 1$:

$$r^{\mathrm{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{1.3}$$

The simple OPE pole ($\partial T$ term) is absorbed entirely and contributes nothing to the collision residue. The even OPE pole at $(z-w)^{-2}$ becomes an odd $r$-matrix pole at $z^{-1}$, and the fourth-order OPE pole becomes a third-order $r$-matrix pole. For a single-generator bosonic algebra, the $d\log$ absorption sends $z^{-2n}$ to $z^{-(2n-1)}$: only odd-order poles survive in the $r$-matrix.

## 2 The closed form

**Computation 2.1** (Virasoro spectral $R$-matrix on primary states)**.** For $\mathrm{Vir}_c$ acting on a primary state $|h\rangle$ of conformal weight $h$, the spectral $R$-matrix is

$$R(z) = z^{2h}\,\exp\Big(-\frac{c}{4z^2}\Big) = z^{2h}\sum_{k=0}^{\infty}\frac{(-c/4)^k}{k!\,z^{2k}}. \tag{2.1}$$

*Proof.* On a primary state $|h\rangle$, the Virasoro modes act as $L_0|h\rangle = h|h\rangle$ and $L_m|h\rangle = 0$ for $m \geq 1$. The collision residue (1.3) has two independent generators on the primary sector: the identity $\mathbf{1}$ (from the central term $c/2$) and $L_0$ (from the linear term $2T$). Since $\mathbf{1}$ and $L_0$ commute on the primary line, the path-ordered exponential reduces to the ordinary exponential:

$$R(z)\,|h\rangle = \exp\left(\oint \sum_{n\geq 0} r_n\, z^{-n-1}\,dz\right)|h\rangle = \exp\Big(2h\log z - \frac{c/2}{2z^2}\Big)\,|h\rangle. \tag{2.2}$$

The $2h\log z$ comes from integrating $2h/z$ (the $L_0$ eigenvalue multiplied by the simple-pole coefficient in the $r$-matrix). The $-c/(4z^2)$ comes from

integrating $-(c/2)/z^3$ along the contour. Exponentiating gives $R(z) = z^{2h}\exp(-c/(4z^2))$. $\square$

The first four series coefficients are

$$R_0 = z^{2h}, \qquad R_2 = -\frac{c}{4}, \qquad R_4 = \frac{c^2}{32}, \qquad R_6 = -\frac{c^3}{384}. \tag{2.3}$$

## 3 Series expansion and bosonic parity

The exponential $\exp(-c/(4z^2))$ is a power series in $z^{-2}$ with alternating signs:

$$\exp\Big(-\frac{c}{4z^2}\Big) = 1 - \frac{c}{4z^2} + \frac{c^2}{32z^4} - \frac{c^3}{384z^6} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\left(\frac{c}{4}\right)^k z^{-2k}. \tag{3.1}$$

Only even powers of $1/z$ appear. This is the bosonic parity constraint: for a single-generator bosonic algebra, the $d\log$ absorption sends the OPE poles at even orders $z^{-2n}$ to odd $r$-matrix poles $z^{-(2n-1)}$, and the path-ordered exponential preserves the parity of the exponent. The $R$-matrix has the form $z^{2h}\cdot f(z^{-2})$ where $f$ is an entire function of $z^{-2}$.

*Remark* 3.1 (Radius of convergence). The series (3.1) converges absolutely for all $z \neq 0$. The $R$-matrix $R(z) = z^{2h}\exp(-c/(4z^2))$ is an entire function of $z^{-2}$ multiplied by the monomial $z^{2h}$. It has an essential singularity at $z = 0$: the non-terminating character of the series is the datum that distinguishes class $M$ from classes $G$, $L$, and $C$, where the corresponding $R$-matrix is polynomial or rational in $z$.

*Remark* 3.2 (The $k$-th coefficient). The general coefficient in the Laurent expansion of $R(z)/z^{2h}$ is

$$[z^{-2k}]\, R(z)/z^{2h} = \frac{(-c/4)^k}{k!}. \tag{3.2}$$

The factorial growth of the denominator ensures absolute convergence; the alternating sign reflects the negativity of the exponent $-c/(4z^2)$ for $c > 0$. For $c < 0$ the signs become uniform and the $R$-matrix grows exponentially as $z \to 0$.

## 4 The $c$-independent identity $S_3(\mathrm{Vir}_c) = 2$

The cubic shadow coefficient $S_3$ measures the first higher obstruction beyond the modular characteristic $\kappa$.

**Theorem 4.1** ($c$-independence of $S_3$)**.** *The cubic shadow coefficient of the Virasoro algebra satisfies*

$$S_3(\mathrm{Vir}_c) = 2 \tag{4.1}$$

*for all values of the central charge $c$.*

*Proof.* The cubic shadow coefficient is the ratio of the linear $T$-mode contribution to the constant central-term contribution in the depth-2 collision residue:

$$S_3 = \frac{(T_{(1)}T)_{\mathrm{lin}}}{(T_{(3)}T)_{\mathrm{const}}}. \tag{4.2}$$

From the OPE (1.2), the relevant modes are $T_{(3)}T = c/2$ (the central term) and $T_{(1)}T = 2T$ (the linear term). The shadow coefficient is

$$S_3 = \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} = 2,$$

where $\kappa(\mathrm{Vir}_c) = c/2$ appears in both numerator and denominator and cancels identically. □

**Corollary 4.2** (Class $M$ membership)**.** *The identity $S_3(\mathrm{Vir}_c) = 2 \neq 0$ certifies that the Virasoro algebra belongs to shadow class $M$ (infinite shadow depth) at every central charge. The critical discriminant $\Delta_{\mathrm{Vir}} = 8\kappa S_4 = 40/(5c+22) \neq 0$ for generic $c$, and the single-line dichotomy forces $r_{\max} = \infty$.*

## 5 Non-formality: the class $M$ signature at genus 0

The closed form $R(z) = z^{2h}\exp(-c/(4z^2))$ encodes the genus-0 non-formality of the Virasoro algebra. The first correction beyond the leading power is

$$R_2 = -\frac{c}{4}, \tag{5.1}$$

which is the coefficient from which the infinite primitive-cumulant series $\Delta_{\mathrm{Vir}}(t) = t^3 + 2t^5 + \cdots$ of the Virasoro shadow obstruction tower descends.

For a chirally Koszul algebra, the $A_\infty$ operations $m_n$ on the bar cohomology $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ encode the non-formality: $m_n = 0$ for $n \geq 3$ is equivalent to formality ($A_\infty$ quasi-isomorphism to the strict associative algebra). The non-vanishing of $R_2$ is the genus-0 obstruction to formality.

**Proposition 5.1** (Non-formality witness)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra with spectral $R$-matrix $R(z) = z^{2h}(1+R_2z^{-2}+R_4z^{-4}+\cdots)$ on a primary line. If $R_2 \neq 0$, then $\mathcal{A}$ is not $A_\infty$-formal on that primary line: the transferred $A_\infty$ operation $m_3 \neq 0$.*

*Proof.* The coefficient $R_2$ is determined by the modular characteristic: on a primary line, $R_2 = -\kappa/2$, which equals $-c/4$ for Virasoro (where $\kappa = c/2$). The shadow-formality identification at arity 3 gives $S_3 \neq 0 \Leftrightarrow m_3 \neq 0$. Since $R_2 \neq 0 \Leftrightarrow S_3 \neq 0$, the non-vanishing of $R_2$ witnesses the failure of $A_\infty$-formality. □

For the Virasoro algebra, $R_2 = -c/4 \neq 0$ for all $c \neq 0$: the Virasoro is non-formal at every nonzero central charge. The non-formality is encoded in the single number $c/4$ and propagates through the exponential to produce infinitely many nonvanishing corrections.

# 6 Comparison with Heisenberg and affine Kac–Moody

The spectral $R$-matrix provides a sharp invariant distinguishing the three main families.

**Example 6.1** (Heisenberg: class $G$)**.** The Heisenberg algebra $\mathcal{H}_k$ has OPE $J(z)J(w) \sim k/(z-w)^2$. The collision residue is $r(z) = k/z$, a simple pole. On the primary sector, $R(z) = z^k$: the $R$-matrix is a monomial. There is no exponential correction; $R_2 = 0$ and $S_3 = 0$. The tower terminates at arity 2: class $G$.

**Example 6.2** (Affine Kac–Moody: class $L$)**.** For $\widehat{\mathfrak{g}}_k$ with Casimir $\Omega$ at level $k$, the collision residue is $r(z) = \Omega/((k+h^\vee)z)$ on the adjoint sector. The $R$-matrix is $R(z) = z^{\Omega/(k+h^\vee)}$, a monomial in $z$ with the Casimir eigenvalue as exponent. The cubic shadow $S_3 \neq 0$ (from the structure constants $f_c^{ab}$), so the tower extends to arity 3; but the quartic and higher terms vanish by independent sum factorization: class $L$.

**Example 6.3** (Virasoro: class $M$)**.** The $R$-matrix $R(z) = z^{2h}\exp(-c/(4z^2))$ has infinitely many nonvanishing coefficients. The essential singularity at $z = 0$ is the datum that no finite truncation of the shadow obstruction tower captures the full $R$-matrix: every arity contributes, and the tower does not terminate.

The comparison is summarized in the following table:

| **Family** | $R(z)$ | $S_3$ | $r_{\max}$ | **Class** |
|---|---|---|---|---|
| Heisenberg $\mathcal{H}_k$ | $z^k$ | $0$ | $2$ | $G$ |
| Affine $\widehat{\mathfrak{g}}_k$ | $z^{\Omega/(k+h^\vee)}$ | $\neq 0$ | $3$ | $L$ |
| Virasoro $\mathrm{Vir}_c$ | $z^{2h}\exp(-c/(4z^2))$ | $2$ | $\infty$ | $M$ |

The progression from monomial (class $G$) through monomial with nontrivial exponent (class $L$) to essential singularity (class $M$) mirrors the increasing complexity of the OPE: quadratic, Lie-type, and non-quadratic respectively.

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